Catalytic QUBO paths for quantum hypergraph partitioning

Baijian Yao[1], Xiwei Pan[2], Wei Zhang[1]

[1]Department of Electronic and Computer Engineering, Hong Kong University of Science and Technology, Hong Kong SAR, China.
[2]Thrust of Advanced Materials, The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China.

Contributing authors: byao@connect.ust.hk; xiwei.pan@connect.hkust-gz.edu.cn; wei.zhang@ust.hk;

Abstract

Quantum hypergraph partitioning requires faithful objectives and useful collective dynamics. We construct temporary QUBO catalysts that preserve the final cut-net problem while shaping its search path. Exact counterexamples expose clique-surrogate distortion; direct HUBO expectations have exponentially weak uncertain-state curvature for large nets. Incidence-dependent couplings control local response, while sparse collective modes supply balanced directions. A classical free-energy machine tests these energies on 30 original hypergraphs with matched balance settings and one shared 48-replica policy. At exact balance, it improves 15 recorded KaHyPar references; at 1% and 3% imbalance it improves none. Substantial quality gaps and preprocessing costs remain. Small coherent simulations test the local catalyst family, and a circuit dataflow workload relates cuts to communication. These results establish endpoint preservation and variational mechanisms, not quantum advantage or an overlap-gap escape.

Keywords: quantum optimization, QUBO, hypergraph partitioning, catalysts, free-energy machine

Introduction

Hypergraph partitioning allocates interacting tasks or components to balanced blocks while limiting the resources shared between them. Unlike an ordinary graph, a hypergraph can represent a signal or data dependency shared by many components as a single net. The cut-net objective charges each crossing net once, making it a natural model for circuit netlists and distributed computations[1,2]. Multilevel methods combine contraction with local refinement[3,4]; KaHyPar and its parallel successor demonstrate the quality and scale available from these classical techniques[5,6]. Bringing quantum optimization to the same problem requires preserving both the many-vertex cost and the balance constraint, rather than only its pairwise approximation.

Quantum annealing searches binary energy landscapes through time-dependent Hamiltonians[7–9]; quantum approximate optimization algorithms (QAOA) use parameterized alternating circuits[10]. Ising formulations map many discrete tasks to these representations[11]. Quadratic unconstrained binary optimization (QUBO) supplies a two-body encoding; higher-order unconstrained binary optimization (HUBO) retains many-body terms. General QUBO modelling

and exact quadratization provide complementary ways to handle objectives and constraints[12–14]. Graph partitioning has been demonstrated on D-Wave hardware and through hybrid QUBO decomposition[15], while constraint-preserving mixers and drivers offer alternative ways to enforce fixed block sizes[16,17]. These advances motivate hypergraph partitioning, but their reported edge-cut results do not directly transfer to a many-pin cut-net objective.

Recent work addresses hypergraphs explicitly. Li et al. derive objective encodings and compare simulated QAOA, annealing and KaHyPar on small 3-uniform instances[18]. Ibrahim et al. optimize distributions over partitions for fairness and variance objectives[19]. The latter tests a different optimization target, so its reported advantages cannot be interpreted as cut-net improvements. Together, these studies expose a gap between quantum-compatible formulations and a scalable search that is competitive on real hypergraphs. A clique reduction can change which partition is optimal, an exact HUBO can have a weak variational response near an uncertain state, and a faithful QUBO can still create barriers through its balance penalty. The problem therefore concerns the search path as well as the endpoint.

A catalyst offers a way to change that path through interactions that vanish when the target Hamiltonian is restored. Quantum annealing studies show that temporary interactions can alter a transition or an energy gap[20–22]. Spin-glass and adiabatic studies also show why a useful interaction must be assessed for its particular model and schedule[23,24]. For hypergraphs, the open question is how incidence structure should determine local and collective interactions and their strength. Overlap-gap theory gives a deeper motivation: for specified random ensembles and algorithm classes, disconnected ranges of near-optimal overlaps obstruct certain stable searches[25–27]. Related bounds apply to shallow local quantum algorithms on specified random problems[28,29]; recent work examines the distinction between quantum and mean-field-inspired searches[30]. This theory does not establish an overlap gap for the real netlists considered here, or imply that a local curvature change removes a global obstruction.

Here we construct incidence-dependent catalytic QUBO paths with an exact cut-net endpoint. We compare objective representations, derive a reduced free-energy Hessian, introduce collective low-frequency couplings, and test partition quality and runtime against KaHyPar. The variational free-energy machine (FEM)[31] supplies a scalable classical testbed for these energies; small coherent simulations separately test the local family's quantum dynamics. A circuit-netlist dataflow experiment then relates partition cuts to measured communication. The results identify useful variational mechanisms and selected competitive partitions, alongside remaining quality gaps and gains already explained by classical spectral information. They establish neither quantum advantage nor an overlap-gap escape.

Results

Objective encodings

Let $x_i \in \{0,1\}$ denote the block of vertex i, and let k_e count ones in net e of size r_e. The desired cost is $E_{\text{cut}} = \sum_e w_e \mathbb{I}(0 < k_e < r_e)$. Two common quadratic surrogates sum $w_e k_e(r_e - k_e)$ or divide that quantity by $r_e - 1$. Both count pairs separated within each net. Even normalizing the pairs cannot make every crossing configuration of an arbitrary net cost exactly one. The difference affects the optimum, not just the quality of a heuristic.

Exact enumeration gives concrete counterexamples (Fig. 1). Each has 12 vertices, ten mixed-size nets and 924 balanced assignments. In example A, the native optimum is 7, but every minimum of the unit-pair surrogate has native cut 8. In example B, the native optimum is 5; the normalized surrogate selects native cut 6, while the unit-pair surrogate has minimum-state cuts ranging from 8 to 10. These are illustrative counterexamples, not estimates of average surrogate performance. Exact HUBO, the auxiliary QUBO and single-run strict KaHyPar all recover the native minima. KaHyPar takes 4.7 ms on each example.

The exact HUBO writes each net as $1 - \prod_{i\in e} x_i - \prod_{i\in e}(1 - x_i)$. It preserves the objective without auxiliaries, but requires interactions up to the net size. Under independent Bernoulli marginals, its expected cost has off-diagonal vertex curvature $-w_e 2^{3-r_e}$ at $p_i = 1/2$. Thus, the structural response around the uncertain state decays exponentially with net size. This is a local

property of that variational representation, not a proof that all HUBO solvers fail. Table 1 separates objective distortion, interaction order and auxiliary cost. The complete QUBO retains the exact endpoint using two auxiliaries per net, and permits a temporary quadratic structural response. Its exact-balance penalty also creates a coordinate barrier: every balanced, auxiliary-minimized binary partition is a one-bit local minimum when $\lambda > D_{\max}$, regardless of its cut. Faithfulness alone therefore cannot supply a useful search dynamic (Methods).

QUBO catalysts

Our approach keeps the complete exact QUBO and modifies its intermediate path. A pair term $\gamma(t)C_{\text{pair}}$ redistributes interactions within nets, while an auxiliary term favors polarized auxiliary states:

$$\begin{aligned} C_{\text{pair}} &= \sum_e w_e a_e \sum_{i<j\in e} z_i z_j (x_i - x_j)^2, \\ C_{\text{aux}}(t) &= -\sum_e \kappa_e(t)(u_e - v_e)^2. \end{aligned} \tag{1}$$

Here u_e, v_e are the two binary auxiliaries for net e. The scales z_i depend on incidence degree and local degree contrast; a_e preserves total coupling mass when these scales change. Both catalysts vanish at the endpoints. Intermediate per-spin temperatures supply another variational control, with their offsets also vanishing before the final original-QUBO phase (Fig. 2). Consequently the final binary objective is unchanged, and a penalty $\lambda > D_{\max}$ guarantees exactly balanced global minima (Methods). This guarantee concerns the encoding, not the output of a finite heuristic.

The graph-science mechanism follows from the reduced free energy after minimizing each four-state auxiliary block. For even n, at uniform temperature and vertex probabilities, the Hessian is

$$\begin{aligned} \mathbf{H} = {} & 4T\mathbf{I} + 2\lambda(\mathbf{1}\mathbf{1}^{\mathsf{T}} - \mathbf{I}) \\ & - \mathbf{B}^{\mathsf{T}} \operatorname{diag}(w_e^2 \chi_e / T)\mathbf{B} \\ & - 2\gamma \mathbf{A}_{\text{off}}, \end{aligned} \tag{2}$$

where $\mathbf{B}$ is incidence, χ_e is the auxiliary susceptibility and $\mathbf{A}_{\text{off}}$ is the graph-dependent off-diagonal pair operator. Projection onto the zero-mean subspace isolates balanced fluctuations. Entropy stabilizes uncertainty; auxiliary susceptibility and catalytic geometry shift the unstable directions. Degree alone cannot determine that operator. The exact curvature expression agrees with automatic second derivatives on weighted test instances. Its minimum eigenvalue changes systematically with catalyst strength on the illustrated hypergraph (Fig. 2(c)).

Local incidence weights need not supply an extended direction across a large graph. We therefore also use zero-mean, unit-norm low-frequency modes $\mathbf{v}_\ell$ of incidence-derived Laplacians, with both unscaled and degree-normalized geometry. Spectral partitioning supplies the classical structural motivation[32,33]. The modes are computed from the original incidence. Their collective quadratic catalyst is

$$\begin{aligned} C_{\text{col}}(t) &= -\frac{\rho(t)}{2} \sum_\ell \theta_\ell \left[(\mathbf{v}_\ell^{\mathsf{T}} \mathbf{s})^2 - \|\mathbf{v}_\ell\|^2 \right], \\ s_i &= 2x_i - 1. \end{aligned} \tag{3}$$

with $\theta_\ell \geq 0$ and $\sum_\ell \theta_\ell = 1$. For one mode, its expected-energy Hessian adds $-4\rho(\mathbf{v}\mathbf{v}^{\mathsf{T}} - \operatorname{diag}(v_i^2))$. The mean-balance rank-one term annihilates $\mathbf{v}$, while this catalyst changes curvature along $\mathbf{v}$ by $-4\rho(1 - \sum_i v_i^4)$. Thus an extended mode supplies a strong balanced instability; a mode localized on one vertex supplies none. This statement concerns local variational curvature, not a quantum gap.

Vertices incident only to singleton nets have zero cut response, but still enter balance. We retain these spins and add a temporary quadratic coupling on a zero-sum random signed direction supported on them. This null-support catalyst encourages their balanced polarization without supplying a binary partition or repairing any vertex. Both collective coefficients vanish at the endpoints. A shared replica portfolio combines individual modes, mode subspaces, local catalysts and per-spin annealing. Component contrasts handle disconnected cut-active supports (Methods).

The overlap-gap picture concerns pairs of near-optimal partitions: their absolute spin overlaps can occupy separated intervals, with no admissible pairs in between (Fig. 2(d)). This is a restriction on the geometry of low-cost solutions; it does not require a flat surface or vanishing gradients. Its algorithmic consequences depend on the problem

Table 1 Cut-net encodings

Construction	Order	Binary variables	Property relevant to cut-net search
Unit-pair clique	2	n	Different objective; example A selects cut 8 instead of 7.
Normalized clique	2	n	Different objective; example B selects cut 6 instead of 5.
Exact HUBO	$\leq r_{\max}$	n	Exact cost; uniform-state response decays as 2^{3-r_e}.
Complete auxiliary QUBO	2	$n+2m$	Exact cost after auxiliary minimization; extra spins and couplings.

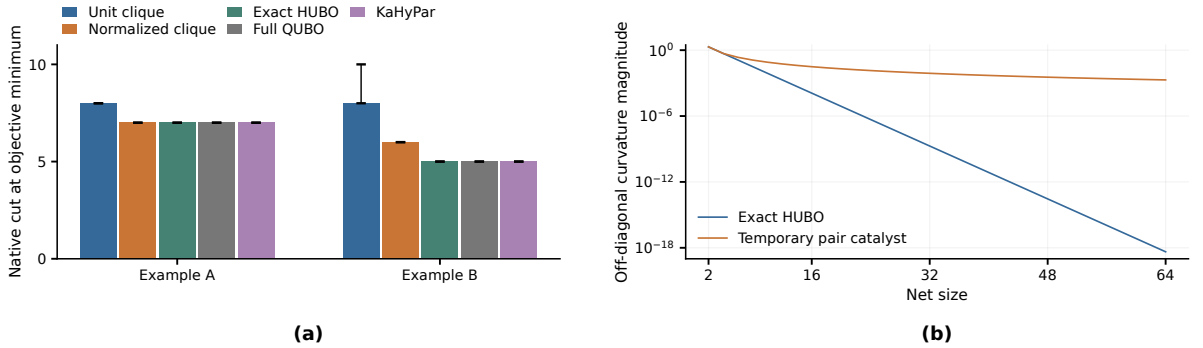


Figure 1 Objective fidelity and curvature; (a) Native cut among global minima of each objective on two exact counterexamples; Bar heights use the best native cut among tied minima; the whisker on example B's unit clique spans all native cuts among its minima, not an uncertainty interval; Exact HUBO and full QUBO recover the same native optima as strict KaHyPar; **(b)** Magnitudes of one net's off-diagonal curvature at independent $p_i = 1/2$: 2^{3-r} for the exact HUBO expectation, and $2/\max(1, \lfloor r^2/4 \rfloor)$ for a unit-strength normalized pair catalyst; The latter is temporary and is not substituted for the final cut objective

ensemble and algorithm class. OGP has not been established on these inputs. A schematic landscape illustrates local trapping and a temporary change in the search direction (Fig. 2(e), (f)), without claiming that catalysts remove an overlap gap. The measured mechanism is the balanced curvature in Fig. 2(c); a computed variational slice is provided in Supplementary Information. The endpoint landscape is restored when the temporary terms vanish.

These terms provide controllable mechanisms without changing the endpoint. They do not prove global optimization or a larger quantum gap. The same quadratic terms define diagonal Ising operators in a transverse-field quantum path (Methods). FEM provides the scalable variational tool, while coherent statevector evolution tests the local pair/auxiliary family on small encodings. The collective extension has not yet been tested in coherent evolution.

The same vertex-only catalytic policy also acts on the native HUBO, without net auxiliaries. On three seen instances, matched 48-replica comparisons improve both representations at exact balance: QUBO cuts change from 516 to 146 on *finan512*, from 5,150 to 2,444 on *IBM13* and from 722 to 209 on *rajat26*; HUBO cuts change from 7,445 to 146, from 4,755 to 2,385 and from 345 to 181, respectively. These comparisons disable QUBO auxiliary polarization and share vertex fields, balance updates and global schedules. The off arms remove both catalytic couplings and local-temperature offsets, so the measured effect concerns their combined search policy. The gain therefore does not require quadratization, although the two variational landscapes remain different. Gradient traces and matched controls are provided in Supplementary Information; these results do not establish coherent many-body dynamics or an overlap-gap escape.

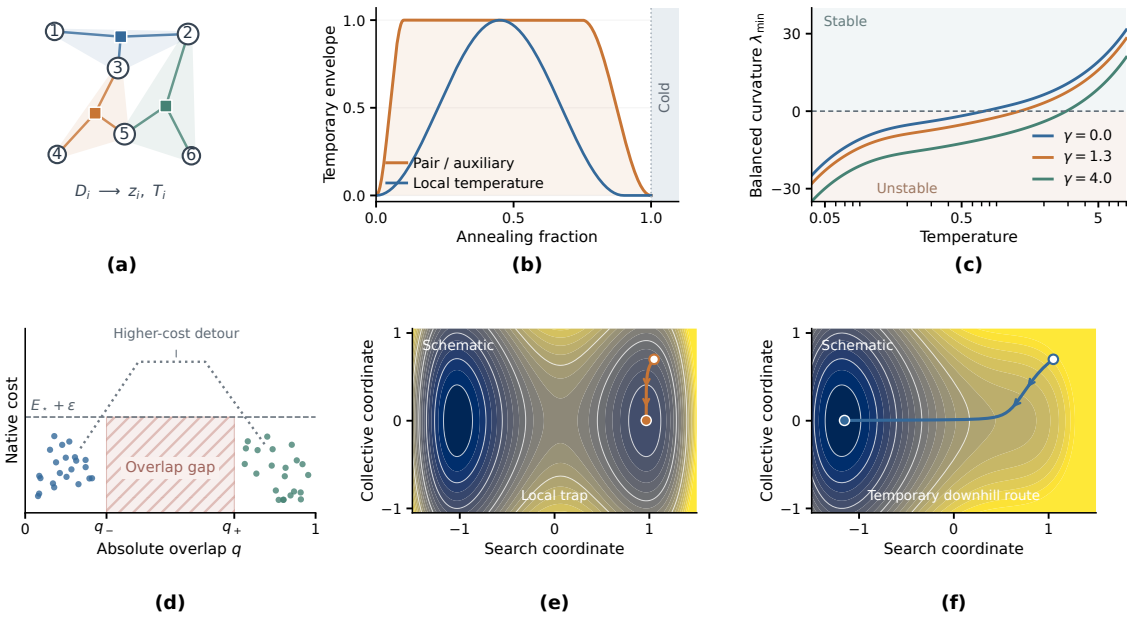


Figure 2 Catalyst landscapes; (a) A schematic shared-net hypergraph; squares represent nets and circles represent vertices, with larger circles indicating higher degree; Degree and local incidence contrast determine structural scales and temperature offsets; **(b)** Temporary pair/auxiliary envelopes and local-temperature offsets vanish before the cold phase; **(c)** Minimum balanced-subspace curvature from Equation (2) on example B, with penalty 0.01 and pair strengths 0, 1.3 and 4; Zero marks the local stability boundary; The curves are classical free-energy curvatures, not quantum spectral gaps; **(d)** Schematic OGP: near-optimal pairs have absolute overlap outside (q_-, q_+); the dotted detour leaves the near-optimal set; **(e)** An illustrative two-basin potential traps gradient descent in a local minimum; **(f)** A temporary term changes the descent direction and removes that local trap in the illustration; Contours indicate increasing potential; open circles mark the common initial state and filled circles the endpoints; Panels (d)–(f) are schematic, not measurements on benchmark graphs; The catalyst vanishes at the end of a search, restoring the original landscape; no OGP escape or measured barrier reduction is inferred

Partition quality

We test 30 original hypergraphs at allowed imbalance 0%, 1% and 3%, using a separate 48-replica batch at each setting. The same local, unscaled and degree-normalized groups receive matched nearby random fields and 12,144 updates. All vertices and net auxiliaries remain in the QUBO; the 1% and 3% encodings add bounded binary slack (Methods). No classical partition initializes the search. Table 2 reports native cut with measured imbalance (%) in parentheses. Each FEM setting is compared with KaHyPar at the same allowed capacity. Bold marks the lower cut or shorter charged time within each matched pair, including ties. The recorded KaHyPar outputs are references, not asserted optima.

The policy improves 15 of 30 recorded 0% references, 0 of 30 1% references and 0 of 30 3% references. There are 4 ties at 1% and 4 at 3%. At exact balance, cuts include 1,171 on *IBM03*, 621 on *IBM04*, 2,443 on *IBM12*, 2,907 on *IBM16* and 2,064 on *cfd1*. Remaining failures are substantial: *IBM13* gives 2,544 against 1,204, and *scircuit* gives 318 against 68. A shared rank-four collective construction therefore does not consistently match multilevel refinement. Allowing more imbalance does not guarantee a lower heuristic result: *IBM01* changes from 325 to 335, and *cfd1* from 2,064 to 2,250. These are independent searches with equal trajectory counts; no strict result is inserted into either band column after seeing its outcome.

The exact-balance policy retains cuts 146 on *finan512*, 255 on *thermal1*, 624 on *G2_circuit* and 1,026 on *parabolic_fem*. KaHyPar's 3% runs on *finan512*, *G2_circuit* and *parabolic_fem* also return exactly balanced partitions, with cuts 146, 624 and 1,026. These remain feasible classical ties even when the separately configured 0% references

Table 2 QUBO cut and measured imbalance;

	Allowed imbalance 0%		Allowed imbalance 1%		Allowed imbalance 3%	
Graph	FEM	KaHyPar	FEM	KaHyPar	FEM	KaHyPar
IBM01	325 (0.000)	**287** (0.000)	368 (0.988)	**220** (0.471)	335 (2.964)	**202** (2.760)
IBM02	477 (0.000)	**367** (0.000)	454 (0.990)	**350** (0.520)	420 (3.000)	**337** (2.989)
IBM03	**1,171** (0.000)	1,247 (0.000)	1,160 (0.994)	**962** (0.000)	1,143 (2.939)	**957** (0.199)
IBM04	**621** (0.000)	696 (0.000)	636 (0.996)	**598** (0.996)	609 (2.995)	**582** (1.898)
IBM05	1,901 (0.000)	**1,807** (0.000)	1,896 (0.402)	**1,725** (0.000)	1,895 (0.450)	**1,719** (1.036)
IBM06	**1,080** (0.000)	1,335 (0.000)	1,078 (0.560)	**1,023** (0.923)	1,080 (0.892)	**988** (1.994)
IBM07	**1,163** (0.000)	1,428 (0.000)	1,144 (0.997)	**948** (0.997)	1,206 (2.996)	**910** (2.992)
IBM08	1,238 (0.000)	**1,200** (0.000)	1,181 (0.998)	**1,140** (0.000)	1,211 (2.276)	**1,140** (0.000)
IBM09	756 (0.000)	**703** (0.000)	723 (0.996)	**627** (0.974)	685 (2.996)	**620** (2.663)
IBM10	2,070 (0.000)	**1,626** (0.000)	1,732 (0.746)	**1,340** (0.000)	1,720 (0.668)	**1,333** (0.349)
IBM11	**1,581** (0.000)	1,671 (0.000)	1,458 (0.998)	**1,079** (0.989)	1,417 (2.999)	**1,062** (2.891)
IBM12	**2,443** (0.000)	3,138 (0.000)	2,457 (0.788)	**1,996** (0.979)	2,492 (2.895)	**1,936** (2.476)
IBM13	2,544 (0.000)	**1,204** (0.000)	2,508 (1.000)	**848** (0.404)	1,921 (3.000)	**848** (0.404)
IBM14	3,259 (0.000)	**2,273** (0.000)	3,088 (1.000)	**1,849** (0.000)	3,577 (3.000)	**1,849** (0.000)
IBM15	4,573 (0.000)	**4,554** (0.000)	4,721 (0.999)	**2,839** (0.722)	4,145 (0.586)	**2,741** (1.868)
IBM16	**2,907** (0.000)	4,044 (0.000)	3,770 (1.000)	**2,117** (0.832)	3,544 (2.746)	**1,969** (2.941)
IBM17	3,446 (0.000)	**3,390** (0.000)	3,366 (0.999)	**2,351** (0.965)	2,784 (3.000)	**2,310** (2.600)
IBM18	**2,197** (0.000)	2,772 (0.000)	2,231 (0.537)	**1,961** (0.000)	2,326 (2.989)	**1,915** (1.355)
2cubes_sphere	**4,125** (0.000)	6,892 (0.000)	3,991 (0.989)	**3,922** (0.993)	3,951 (2.999)	**3,884** (2.993)
G2_circuit	**624** (0.000)	1,144 (0.000)	**624** (0.779)	**624** (0.000)	**624** (0.779)	**624** (0.000)
PGPgiantcompo	463 (0.000)	**396** (0.000)	467 (0.449)	**352** (0.000)	457 (2.959)	**350** (0.000)
airfoil_2d	375 (0.000)	**372** (0.000)	373 (0.999)	**328** (0.000)	342 (2.997)	**324** (2.899)
bcsstk29	372 (0.000)	**360** (0.000)	**360** (0.172)	**360** (0.000)	**360** (0.172)	**360** (0.000)
ca-CondMat	4,473 (0.000)	**4,202** (0.000)	4,406 (0.994)	**3,918** (0.916)	4,215 (3.000)	**3,855** (3.000)
cfd1	**2,064** (0.000)	3,121 (0.000)	2,174 (0.999)	**1,564** (0.725)	2,250 (2.998)	**1,564** (0.725)
finan512	**146** (0.000)	169 (0.000)	**146** (0.000)	**146** (0.000)	**146** (0.000)	**146** (0.000)
parabolic_fem	**1,026** (0.000)	1,256 (0.000)	**1,026** (0.013)	**1,026** (0.000)	**1,026** (0.013)	**1,026** (0.000)
scircuit	318 (0.000)	**68** (0.000)	277 (0.999)	**46** (0.513)	46 (2.451)	**42** (2.273)
shallow_water2	**768** (0.000)	909 (0.000)	768 (0.764)	**640** (0.000)	768 (0.601)	**640** (0.000)
thermal1	**255** (0.000)	291 (0.000)	257 (0.102)	**239** (0.111)	257 (0.099)	**239** (0.114)

are worse. Classical rounding of eight continuous incidence modes matches these three cuts and reaches 272 on *thermal1*; FEM improves that last cut by 17 nets. Spectral information alone explains several strong outcomes, and its rounded partitions are assessed separately.

The FEM score includes the specified original-QUBO one-bit descent, whose time is charged for all replicas. On 29 exact-balance instances, the winning raw partition is unchanged. On *scircuit*, the best feasible raw cut is 373; a different raw replica with count residual 2 reaches cut 318 through two vertex and four auxiliary flips. This exception is reported explicitly. Raw feasibility counts, cut differences and winner flip counts for all three balance settings are in Supplementary Information.

Table 3 separates timing from solution quality. The FEM cost includes both mode constructions, graph setup, compilation/capture, all 48 trajectories, checkpoints and all-replica descent. CPU preprocessing costs are summed even when jobs overlap, so this is a conservative charged cost. On *parabolic_fem*, the original CPU modes cost 1466.81 s and the strict GPU loop costs 467.52 s. On *scircuit*, an unsuccessful CPU mode attempt and a certified A100 replacement are both charged. Separate profiling identifies slow mode construction and sparse products; it does not replace the recorded solver timings. Hardware, reference budgets and preprocessing execution differ, so no equal-budget speedup is inferred.

Twenty inputs informed earlier studies. Ten additional inputs extend the fixed policy to the full *IBM01–IBM18* family and three further matrix-derived hypergraphs. One base seed

Table 3 QUBO charged time (s);

	Allowed imbalance 0%		Allowed imbalance 1%		Allowed imbalance 3%	
Graph	FEM	KaHyPar	FEM	KaHyPar	FEM	KaHyPar
IBM01	**25.35**	128.68	**20.73**	64.32	**21.98**	64.67
IBM02	**31.42**	128.30	**27.52**	64.61	**29.46**	64.44
IBM03	**33.19**	128.61	**29.23**	64.15	**29.79**	64.51
IBM04	**34.91**	128.67	**31.20**	64.23	**33.00**	65.10
IBM05	**35.75**	128.50	**32.00**	64.26	**33.43**	64.30
IBM06	**37.31**	129.22	**34.35**	64.66	**35.32**	66.49
IBM07	**49.79**	128.73	**48.05**	64.40	**49.88**	65.25
IBM08	**105.68**	128.37	103.51	**64.82**	105.71	**64.52**
IBM09	**59.74**	128.67	**58.02**	64.48	**59.28**	64.66
IBM10	**84.20**	129.23	82.91	**64.43**	85.31	**64.85**
IBM11	**78.65**	128.55	77.93	**64.41**	80.82	**64.90**
IBM12	**91.24**	128.81	90.88	**66.19**	91.72	**66.35**
IBM13	**103.27**	130.33	103.87	**65.48**	106.56	**64.40**
IBM14	212.18	**128.97**	219.53	**66.70**	220.02	**65.71**
IBM15	208.97	**65.63**	214.57	**64.59**	216.81	**69.89**
IBM16	297.77	**136.65**	304.09	**68.51**	307.32	**72.02**
IBM17	215.52	**144.28**	221.60	**66.23**	223.75	**65.19**
IBM18	252.76	**130.88**	260.83	**67.02**	263.85	**67.20**
2cubes_sphere	132.94	**129.14**	134.23	**67.97**	**134.80**	155.80
G2_circuit	163.52	**129.09**	170.21	**64.91**	173.42	**67.27**
PGPgiantcompo	**24.87**	128.57	**20.57**	64.31	**21.61**	64.40
airfoil_2d	**28.33**	128.35	**24.24**	64.16	**24.81**	64.23
bcsstk29	**32.47**	129.15	**28.11**	64.27	**29.45**	64.41
ca-CondMat	**40.99**	128.85	**36.71**	65.58	**37.52**	66.07
cfd1	**94.29**	130.68	93.73	**64.89**	95.12	**64.73**
finan512	**86.46**	128.59	86.11	**64.52**	88.30	**64.52**
parabolic_fem	1971.66	**133.58**	2004.51	**67.59**	2007.91	**80.87**
scircuit	3352.19	**132.23**	3419.80	**65.11**	3744.58	**66.07**
shallow_water2	**74.41**	128.99	73.75	**64.73**	75.30	**64.85**
thermal1	**106.93**	128.47	105.68	**64.90**	107.03	**66.02**

per graph does not establish seed robustness or fully unseen generalization. The native cut objective and scale differ from small 3-uniform QAOA studies[18], distributional fairness/variance optimization[19], and D-Wave graph partitioning through decomposition[15]. Full group comparisons and exploratory controls remain in Supplementary Information.

Tables 4 and 5 report the separate HUBO comparison on three seen inputs, all at 0% measured imbalance. The shared vertex policy is compared with a fixed mixture of four timing paths within the same 48-replica budget. The mixed policy reaches 2,232 on *IBM13* and 116 on *rajat26*, while retaining 146 on *finan512*. It improves the shared HUBO policy on two inputs, but the corresponding QUBO mixture regresses on *rajat26* and improves *IBM13* to 1,941. No better old/new result is substituted into the 30-input QUBO table. Both encodings remain well above KaHyPar on the first two inputs. These controls support a useful many-body catalytic path on selected cases; broad HUBO transfer and additional balance settings remain untested.

The separate coherent test uses eight six-vertex, 14-qubit complete encodings. At duration 12, mean probability of an optimal balanced vertex partition rises from 23.02% to 37.04%, improving on all eight cases. Full-QUBO ground-state probability rises from 2.31% to 3.12% on average, with one regression. Mean balance probability falls from 85.77% to 83.73% (Fig. 3). Target-partition probability and encoded state success

Table 4 HUBO cut and measured imbalance;

Graph	Off	Shared catalyst	Mixed catalyst	KaHyPar
finan512	7,445 (0.000)	**146** (0.000)	**146** (0.000)	169 (0.000)
IBM13	4,755 (0.000)	2,385 (0.000)	2,232 (0.000)	**1,204** (0.000)
rajat26	345 (0.000)	181 (0.000)	116 (0.000)	**32** (0.000)

Table 5 HUBO charged time (s);

Graph	Off	Shared catalyst	Mixed catalyst	KaHyPar
finan512	**75.62**	112.44	113.77	128.59
IBM13	**88.97**	138.38	139.20	130.33
rajat26	**48.85**	93.05	93.48	128.54

are therefore different observables. Strict KaHyPar reaches the exact cut on all eight inputs in 2–4 ms. These classical simulations of quantum evolution demonstrate a catalyst effect, not quantum advantage.

Circuit dataflow

To apply the partitions, we construct a two-worker synchronous dataflow workload on *IBM01*, *IBM06* and *IBM15* connectivity. Each vertex has a numerical state; each net reduces its incident states and distributes that value to its local operators. Crossing nets exchange contributions between workers. Net payloads contain 8 or 32 double-precision values. The connectivity is from real circuit netlists, while operator functions are defined by the benchmark. These files lack signal directions and gate functions, so the workload is not gate-level circuit simulation.

For payload width d, the transmitted volume per step is exactly $V = 2d\, s_{\mathrm{fp64}} E_{\mathrm{cut}}$, with $s_{\mathrm{fp64}} = 8$ bytes. Instrumented two-process messages agree with this count in all 54 trials. All parallel outputs agree with the serial computation to within 1.12×10^{-16} in maximum absolute error. This establishes an application-level meaning for the cut-net objective: each crossing shared dependency contributes one fixed bidirectional payload, irrespective of its number of pins.

The application retains both base-seed partitions from the fixed portfolio and the strict KaHyPar partition. Averaging the two FEM seed volumes gives ratios to KaHyPar of 0.9948 on *IBM01*, 1.0210 on *IBM06* and 1.0890 on *IBM15*. The best *IBM15* seed matches KaHyPar's 4,554 crossing nets; the other has 5,365. No application result is selected after timing. At width 32, measured per-step medians are 5.63–5.84 ms for the *IBM01* FEM partitions versus 5.74 ms for KaHyPar, and 110.35–112.80 ms for *IBM15* versus 121.77 ms (Fig. 4). Three repeats per partition and payload are retained. These are observed local CPU microbenchmark times; communication volume, sparse-work distribution and process overhead jointly affect latency, so a cut reduction alone does not guarantee application acceleration.

Discussion

A faithful quantum encoding and a useful search mechanism are separate requirements. Permanent clique substitution can change the desired partition even at its own optimum. Exact HUBO avoids that distortion, but its independent-marginal response becomes weak around uncertain large nets. Temporary quadratic interactions retain the exact endpoint and supply graph-dependent curvature. The incidence-based analysis explains this local control; the coherent experiment shows that its observable consequences also depend on feasibility and auxiliary readout.

The large-instance results show that resolved collective directions can close quality gaps left by local incidence weights. Null-support polarization additionally resolves balance ambiguity on *thermal1*. However, simple spectral rounding and exactly balanced KaHyPar outputs match the solver on *finan512*, *G2_circuit* and *parabolic_fem*. *IBM02*, *PGPgiantcompo* and *ca-CondMat* remain above KaHyPar, and the matched 1% and 3% results show substantial remaining quality gaps.

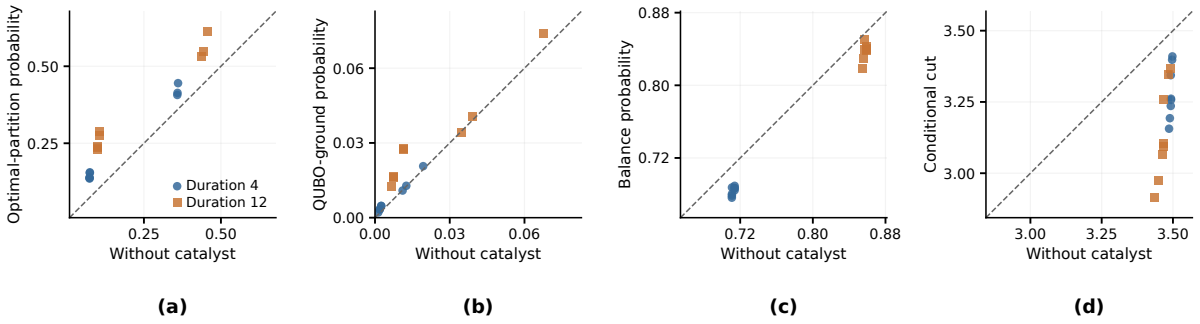


Figure 3 Coherent annealing; Eight matched 14-qubit encodings are simulated at durations 4 and 12; Horizontal axes show uncatalyzed observables and vertical axes show catalyzed observables; dashed lines indicate equality; **(a)** Optimal balanced vertex-partition probability, summed over auxiliary bits; **(b)** Full-QUBO ground-state probability; **(c)** Balance probability; **(d)** Native cut conditional on balance; Higher probability is preferred in (a)–(c) and lower cut in (d); Exact enumeration and strict KaHyPar agree on cuts 2 or 3 on every instance

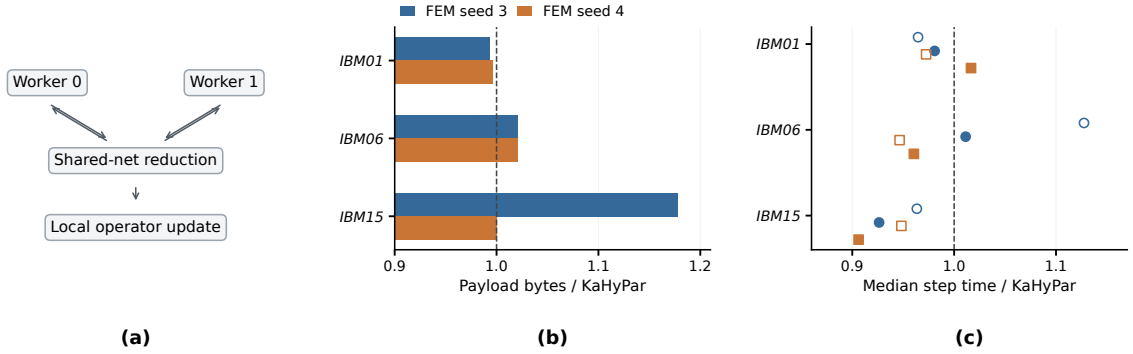


Figure 4 Circuit dataflow; (a) Shared-net reduction/broadcast and local numerical updates on two workers; **(b)** Measured transmitted-payload ratios to the strict KaHyPar partition at width 32; Both fixed FEM base seeds are shown; **(c)** Ratios of per-step median runtime from three repeats; Circles mark seed 3 and squares seed 4; open markers use width 8 and filled markers width 32; Values below one improve the corresponding KaHyPar application metric; The experiment measures two local CPU workers with synthetic operator functions on real circuit connectivity, not FPGA or gate-level simulation; Full cut/partition-time and application-time tables are in the Supplementary Information

These comparisons limit both the algorithmic and quantum interpretation: useful QUBO search paths are demonstrated, while broad competitiveness and an advantage over classical spectral information remain unproven. The circuit-derived workload connects cut quality to real exchanged bytes, while exposing why task latency also depends on workload distribution. Weighted operator costs, multiple processors and an actual simulator remain necessary before extrapolating to deployed circuit simulation.

OGP provides a theoretical motivation rather than a completed explanation of these real instances. Neither the local Hessian nor overlap among a few returned partitions determines forbidden intervals among all near-optimal states. A global claim requires a specified ensemble, threshold, algorithm class and scaling test. Correlated variational models, including cluster mean-field methods[34], autoregressive variational distributions[35] and tensor-network representations[36], offer possible ways to retain correlations beyond independent vertex marginals. Transverse quantum catalysts supply a different mechanism. These extensions require separate tests beyond the diagonal catalysts evaluated here. The present contribution is an exact-endpoint catalyst framework, local mechanism and reproducible partition/application tests, with its limits stated.

Methods

Objective comparisons

For clique objectives, we evaluate both unit pair weights and weights $1/\max(r_e-1,1)$. Exact enumeration scores every balanced assignment on 12 vertices. Fixed generation seeds 200–1199 define ten mixed-size nets; the first strict counterexample for each surrogate is retained, giving seeds 200 and 207 after eight examined graphs. All surrogate minimizers are included and the best native cut among ties is reported. These are explicitly selected mechanism examples, not a representative benchmark. All 924 assignments per example agree with the independently scored auxiliary QUBO after conditional auxiliary minimization. KaHyPar uses one ordinary seed-0 strict-balance run per example.

The exact HUBO Bernoulli expectation is $\sum_e w_e[1-\prod_{i\in e}p_i-\prod_{i\in e}(1-p_i)]$. Differentiating twice at independent $p_i=1/2$ gives zero diagonal curvature and off-diagonal $-w_e 2^{3-r_e}$ within a net. The pair catalyst has off-diagonal $-2\gamma w_e a_e z_i z_j$. For Fig. 1(b), weights/scales and strength equal one, and $a_e=1/\max(1,\lfloor r_e^2/4\rfloor)$. Fig. 2(c) uses example B, a penalty of 0.01 and 100 logarithmically spaced temperatures from 0.04 to 8. Auxiliary polarization is zero; exact balanced-subspace eigenvalues are computed from the verified matrix-free Hessian.

Cut-net QUBO

For nonnegative net weights, the objective is $E_{\mathrm{cut}}(\mathbf{x})=\sum_e w_e\mathbb{I}(0<k_e<r_e)$ with $\sum_i x_i=b$. For odd n, block sizes are b and $n-b$. Two auxiliaries per nonempty net give

$$\min_{u_e,v_e\in\{0,1\}}\big[1+(r_e-1-k_e)u_e + (k_e-1)v_e\big]=\mathbb{I}(0<k_e<r_e). \tag{4}$$

The complete QUBO multiplies each net term by w_e and adds the balance penalty.

Equation (4) follows by examining the three cases $k_e=0$, $0<k_e<r_e$ and $k_e=r_e$. Minimizing the two auxiliary bits gives costs 0, 1 and 0, respectively. A state with imbalance $d=|\sum_i x_i-b|$ can be balanced by flipping d vertices. Its native cut can increase by at most $dD_{\max}$, whereas removing its penalty saves λd^2. Thus $\lambda>D_{\max}$ excludes imbalanced global minima. The implementation uses $D_{\max}+1$ and retains the complete $n+2m$ binary state for independent scoring.

This equality concerns binary energies after auxiliary minimization; it does not equate the HUBO and QUBO variational landscapes. At a fixed vertex distribution, the expectation of the auxiliary minimum can be smaller than the minimum expected energy of auxiliaries independent of vertices. Conditional auxiliaries that select a minimizing state for each vertex assignment recover the HUBO expectation. Our factorized vertex distribution and within-net auxiliary factors do not impose these vertex–auxiliary correlations. Auxiliary entropy and a quantum driver also affect thermodynamics and dynamics beyond binary endpoint equivalence.

For an allowed imbalance ϵ, let $c=\lceil n/2\rceil$, $U=\lfloor(1+\epsilon)c\rfloor$, $L=n-U$ and $N=\sum_i x_i$. A bounded binary slack $S=\sum_j a_j b_j$ spans all integers from 0 to $U-L$. Its weights are powers of two followed by the remaining range. The matched 1% and 3% QUBOs replace exact balance by

$$E_{\mathrm{bal}}=\lambda(N-L-S)^2. \tag{5}$$

Minimizing slack at fixed vertices gives $\lambda\,\mathrm{dist}(N,[L,U])^2$. The same vertex-flip bound proves that $\lambda>D_{\max}$ excludes infeasible global minima. This uses $\lceil\log_2(U-L+1)\rceil$ extra bits, while retaining every vertex and net auxiliary. No binary vertex is projected or repaired by a balance rule. Reported imbalance is $100[\max(n_0,n_1)/\lceil n/2\rceil-1]$; the smallest possible odd split therefore has zero reported imbalance.

The same coefficient implies an endpoint coordinate barrier. At a balanced state, flipping vertex i while fixing auxiliaries changes energy by

$$\Delta_i E=\lambda+(1-2x_i)\sum_{e\ni i}w_e(v_e-u_e)\geq\lambda-D_i>0. \tag{6}$$

If auxiliaries are conditionally minimized, none of their single flips lowers energy either. Every such balanced state is therefore a one-bit local minimum, including poor cuts. One-bit descent can restore feasibility and auxiliary consistency but cannot improve these already feasible endpoints. This lemma is independently checked by exhaustive weighted small-instance enumeration and on

384 saved *finan512/thermal1* states. It concerns coordinate dynamics of the encoding, not OGP or a limitation on correlated continuous or quantum dynamics. The band encoding has the same one-bit barrier when $N = L + S$: vertex flips pay at least $\lambda - D_i$, and a single slack flip pays λa_j^2. Allowing more feasible block sizes does not remove this encoding-induced barrier.

The initial real-instance study uses unweighted vertices and nets and two blocks. It does not yet model heterogeneous task loads or multiprocessor connectivity costs. Circuit-netlist inputs are ISPD98 instances; other inputs are matrix-derived or network hypergraphs from the local benchmark collection. Exact file hashes define every comparison. The additional dataflow benchmark uses synthetic numerical operators on that circuit connectivity; it does not measure gate-level or FPGA latency.

Variational FEM

Mean-field annealing replaces discrete states by continuous marginals[37]. FEM follows this variational approach[31], with entropy and expected energy defined for an explicit trial distribution[38]. For replica a, the distribution is $q_a(x, u, v) = \prod_i \mathrm{Bern}(x_i; p_i^a) \prod_e q_e^a(u_e, v_e)$. The free-energy objective contains the expected QUBO and catalysts, vertex negative entropy $\sum_i T_i(t)[p_i \ln p_i + (1-p_i)\ln(1-p_i)]$, and auxiliary negative entropy $T(t) \sum_{e,u,v} q_e(u, v) \ln q_e(u, v)$. The four-state auxiliary block is minimized exactly at each vertex update. It retains within-edge auxiliary correlation but no vertex–vertex correlation. For $\bar{k}_e = \sum_{i \in e} p_i$, its nonconstant state energies are 0, $w_e(r_e - 1 - \bar{k}_e) - \kappa_e$, $w_e(\bar{k}_e - 1) - \kappa_e$, and $w_e(r_e - 2)$. The reduced block contribution is $w_e - T \ln \sum_{u,v} \exp[-E_e^{\mathrm{nc}}(u, v)/T]$, where E_e^{nc} denotes these four nonconstant energies. The supplementary variational slice uses $\mathbf{p} = \mathbf{1}/2 + q_1\mathbf{v}_1 + q_2\mathbf{v}_2$; zero-mean orthonormal modes keep the mean balance fixed. Vertex entropy, the full balance expectation and diagonal corrections to the collective expectation are retained. For the schematic landscapes only, a dimensionless illustration uses $V_0(x, y) = (x^2-1)^2 + 0.6y^2 + 0.25x$ and $V_g = V_0 + g[-0.55(x^2 - 1)^2 + 0.6x]$. The displayed paths follow gradient descent from $(1.05, 0.7)$ with step 0.025, at $g = 0$ and $g = 1$. These illustrative coordinates are not spin overlaps. Setting $g = 0$ restores V_0 exactly; this construction is not a hypergraph experiment.

The expected balance cost is

$$\mathbb{E}\left[(\textstyle\sum_i x_i - b)^2\right] = (\textstyle\sum_i p_i - b)^2 + \sum_i p_i(1-p_i). \quad (7)$$

Dropping the variance term would change the variational objective. For the direct HUBO comparison, the trial distribution has only vertex Bernoulli factors and no auxiliary entropy. Its free energy is

$$F_{\mathrm{H}}(\mathbf{p}, t) = \mathbb{E}(E_{\mathrm{cut}} + C_t) + \lambda_t \mathbb{E}[(N - b)^2] + \sum_i T_i(t)\phi(p_i). \quad (8)$$

Here $\phi(p) = p \ln p + (1 - p)\ln(1 - p)$ and the cut expectation uses the native HUBO products given above. The native cut gradient is $g_i = \sum_{e \ni i} w_e[\prod_{j \in e \setminus \{i\}}(1 - p_j) - \prod_{j \in e \setminus \{i\}} p_j]$. Incidence products of log probabilities evaluate these terms stably; singleton contributions cancel exactly. Temporary pair and collective terms retain their quadratic expectations. The controlled QUBO arm uses the same vertex catalysts with auxiliary polarization set to zero; its conditional auxiliary entropy remains present. All direct-encoding comparisons use 64 centrally safeguarded implicit iterations, Torch CSR, native fused dense updates and 32-update capture. They are distinct from the frozen primary QUBO protocol. Full original-polynomial one-bit descent is applied to all replicas of each encoding, with no paired moves or capacity repair. For the 1% and 3% runs, one correlated slack block is independent of the vertex and net blocks. If g_s counts the binary representations of slack value s, conditional minimization gives $q_s \propto g_s \exp[-\lambda_t(\mu - L - s)^2/T]$, where $\mu = \sum_i p_i$. Multiplicity retains the entropy of the underlying binary states. The expected balance term is $\lambda_t[(\mu - L - \mathbb{E}S)^2 + \sum_i p_i(1 - p_i) + \mathrm{Var}(S)]$. The current slack expectation provides the per-replica target in the same implicit vertex update, and is refreshed before each update. Binary readout minimizes slack and net auxiliaries with vertices fixed.

Fields h_i give $p_i = \mathrm{sigmoid}(h_i)$ and $m_i = \tanh(h_i/2)$. The manual update separates the sparse cost gradient g_i from balance. With $d_i = \eta_a/(1 + D_i)$, set $r_i = d_i/(1 + 2\lambda_t d_i)$ and $\rho_i =$

$r_i/(1+r_iT_i)$. The tilted field is $\widetilde{h}_i = (h_i - r_i[g_i - \lambda_t m_i])/(1+r_iT_i)$, clipped to $[-20, 20]$. The new field is $h'_i = \widetilde{h}_i - 2\lambda_t\rho_i\delta$, where

$$\delta = \sum_i \mathrm{sigmoid}(\widetilde{h}_i - 2\lambda_t\rho_i\delta) - b. \tag{9}$$

The corresponding root function is strictly increasing, with derivative $1+\sum_i 2\lambda_t\rho_i p'_i(1-p'_i) > 0$. Twenty-four bracketed Newton iterations solve this scalar equation. It is a soft-penalty update, not a projection onto a balanced binary set.

Local catalysts

For the pair catalyst, the base coefficient is $a_e^{(0)} = 1/\max(1, \lfloor r_e^2/4 \rfloor)$. Node scales take the form $z_i = (1+D_i)^{-\alpha}\exp(\xi s_i)$, where s_i is a bounded structural signal. Coefficients are rescaled so that $\sum_e w_e a_e \sum_{i<j\in e} z_i z_j$ equals its base value. Degree signals use $\ln(1+D_i)$; local contrast subtracts the weighted neighbor mean, counting repeated net co-occurrences. Signals are centered, standardized, clipped to two standard deviations, recentered and bounded; isolated vertices receive zero signal. Incidence operations compute these quantities without constructing a clique adjacency matrix.

Temporary local temperatures are $T_i^a(t) = T(t)\exp[\omega(t)\zeta_a s_i]$, with $\omega(t) = \sin^2(\pi t/0.9)$ for normalized annealing time $0 \le t < 0.9$ and zero thereafter. The pair and auxiliary envelopes rise over the first 10% of annealing and decay over the final 25%, using squared-sine ramps. Auxiliary strengths multiply $w_e \max(r_e/2 - 1, 0)$. The balance penalty remains 0.01 for the first 90% and increases geometrically to $D_{\max}+1$ over the final 10%. In the primary benchmark, temperature decreases exponentially from 8 to 0.02 over 12,000 updates; 144 additional updates retain the final temperature, full penalty and zero catalysts. Fixed checkpoints occur every 128 updates and at both phase boundaries. Selection uses the original full-QUBO energy.

Each 16-replica batch starts from a shared normal field with standard deviation 0.01 plus independent perturbations with standard deviation 0.001. Base seeds determine both fields reproducibly. The frozen portfolio combines negative and positive contrast weighting, degree-temperature biases, local-contrast cooling, inverse-degree geometry, a geometry transition and uniform auxiliary alternatives. The exact profiles and their hashes are retained in the frozen protocol. Replicas evolve as independent variational starts, without the temperature-swap transitions of exchange Monte Carlo[39]. Per-spin temperature offsets are classical variational controls; they do not implement the inhomogeneous transverse-field schedules studied in quantum annealing[40]. No graph name determines a profile or parameter.

Collective catalysts

Let $\mathbf{B}$ be net-by-vertex incidence, $\mathbf{r}$ the net sizes and $a_e = 1/\max(1, \lfloor r_e^2/4 \rfloor)$. The implicit operator is $\mathbf{L} = \mathrm{diag}(\mathbf{B}^\mathsf{T}(\mathbf{a}\odot\mathbf{r})) - \mathbf{B}^\mathsf{T}\mathrm{diag}(\mathbf{a})\mathbf{B}$. Its products use sparse incidence without storing clique edges. Let $\mathbf{D}_c = \mathrm{diag}((L_{ii})_i)$ on the cut-active support, and let $\mathbf{P}_c$ remove the mean separately within each connected component. We also compute modes of

$$\mathbf{L}_{\mathrm{norm}} = \mathbf{P}_c\mathbf{D}_c^{-1/2}\mathbf{L}\mathbf{D}_c^{-1/2}\mathbf{P}_c. \tag{10}$$

These degree-normalized physical directions are not eigenvectors of the unscaled operator. Inactive vertices are excluded only from these eigenproblems and retained in the complete QUBO.

Each geometry supplies four zero-mean, orthonormal physical directions. On connected support we discard the constant mode. On disconnected support, random component contrasts occupy its nonconstant nullspace. If the largest component exceeds half the total vertex count, we keep one contrast and three internal directions; otherwise, up to four contrasts are retained. Internal modes use a shift on the component nullspace. The Lanczos solver uses tolerance 10^{-6}, maximum 20,000 iterations, subspace size 32 and probe seed 314159; absolute residuals are checked below 10^{-5}. Modes are padded with zeros on inactive spins.

For $\mathbf{m} = 2\mathbf{p} - \mathbf{1}$, one catalyst's exact Bernoulli expectation is $-\rho[(\mathbf{v}^\mathsf{T}\mathbf{m})^2 - \sum_i v_i^2 m_i^2]/2$, with gradient $-2\rho[\mathbf{v}(\mathbf{v}^\mathsf{T}\mathbf{m}) - \mathbf{v}^{\odot 2}\odot\mathbf{m}]$. The diagonal is removed because binary $s_i^2 = 1$. Weighted subspaces sum these quadratic expectations. The common 48-replica policy assigns 16 replicas to local catalysts, 16 to unscaled modes and 16 to normalized modes. The latter groups cycle through first mode, second mode, equal first two and normalized inverse-eigenvalue weights over all four. The inverse

uses floor $\max(10^{-12}, 10^{-6}\max_\ell \mu_\ell)$ to handle null modes. These choices cross with strengths $\{0.5, 1, 2, 4\}\max(1, \bar{D})$. The local, auxiliary and per-spin profiles act simultaneously in every group. All three groups receive identical nearby initial fields.

For $f > 1$ inactive spins, each null direction assigns $1/\lfloor f/2 \rfloor$ to one random subset and $-1/\lceil f/2 \rceil$ to its complement, then normalizes. Thus it has zero mean for odd or even support sizes; one inactive spin cannot supply such a direction. Thermal1 has 248 inactive spins and *ca-CondMat* has 610. Fixed replica-dependent seeds match the two collective groups. Local replicas have no null coupling. The collective envelope rises over annealing fractions 0–0.15 and falls over 0.80–1; the null envelope rises over 0.55–0.70 and falls over 0.96–1. Squared-sine ramps give zero endpoints. Null strengths multiply $D_{\max} + 1$ by $\{0.5, 1, 2, 4\}$.

Temperature scheduling follows classical annealing[41]. All replicas use 12,000 exponential updates from 8 to 0.02 and 144 cold updates, with the same final penalty ramp. Vertex signs provide raw readout, followed by auxiliary-only minimization. Checkpoints every 128 updates and at phase ends retain the minimum original-QUBO energy per replica. Separate classical references stably sort each of the eight modes and assign exactly $\lfloor n/2 \rfloor$ vertices to one block. These binary partitions never enter FEM.

Pair correlations

The supplementary control replaces independent vertices by full-support four-state pair factors, retaining every original QUBO variable. With S_g the occupancy of factor g, balance variance is $\sum_g \mathrm{Var}(S_g)$, including within-pair covariance. The quadratic catalyst receives its corresponding covariance correction. Joint entropy at unequal vertex temperatures uses $T_i H_-(p_i) + T_j H_-(p_j) + \min(T_i, T_j)\,\mathrm{I}(x_i; x_j)$. Here H_- is binary negative entropy and I mutual information. The implicit root uses 40 iterations, and the singleton limit reproduces the primary independent equations when clipping is inactive. Joint log weights span at most 80, versus the primary tilted-field bound 20. Output uses factor modes followed by auxiliary-only minimization; no vertex repair is applied. Predetermined factor refresh preserves marginals but discards covariance, so its expected energy jumps are recorded rather than assumed to vanish. It is a classical variational restart, not a physical catalytic-state invariant.

Local stability

For even n, uniform vertex probabilities are a stationary point of the reduced block-variational objective when the catalyst has no linear field. At uniform temperature, Eq. (2) gives its vertex-probability Hessian, where $\mathbf{A}_{\mathrm{off}}$ contains the off-diagonal pair weights $\sum_{e \ni i,j} w_e a_e z_i z_j$, and

$$\chi_e = \left[1 + \exp(-\kappa_e/T)\cosh\!\big(w_e(r_e/2 - 1)/T\big)\right]^{-1}. \tag{11}$$

Indeed, $\chi_e = \mathrm{Var}_{q_e}(v_e - u_e)$ at the uniform state; minimizing the auxiliary Gibbs block contributes its negative susceptibility to the Hessian. The entropy Hessian is $4T\mathbf{I}$, and the exact balance expectation contributes $2\lambda(\mathbf{1}\mathbf{1}^\mathsf{T} - \mathbf{I})$. The relevant balanced-subspace operator is $\mathbf{PHP}$, with $\mathbf{P} = \mathbf{I} - \mathbf{1}\mathbf{1}^\mathsf{T}/n$. Per-spin temperatures replace the entropy term by $4\,\mathrm{diag}(T_i)$. This expression is checked against automatic second derivatives of the complete reduced free energy on weighted small hypergraphs, with and without auxiliary polarization. Matrix-free incidence products evaluate the operator without storing a clique matrix. Its eigenvalues are local classical free-energy curvatures, not the eigenvalue gaps of Eq. (12).

Quantum Hamiltonian

Binary x_i is represented by $\widehat{\mathbf{x}}_i = (\mathbf{I} - \widehat{\mathbf{Z}}_i)/2$, where $\widehat{\mathbf{X}}_i$ and $\widehat{\mathbf{Z}}_i$ are Pauli operators; the QUBO and temporary terms become diagonal Ising operators. A transverse-field path is

$$\begin{aligned} \widehat{\mathbf{H}}(s) = -(1-s)\sum_j \widehat{\mathbf{X}}_j \\ + s\widehat{\mathbf{E}}_{\mathrm{QUBO}} + f(s)\widehat{\mathbf{C}}, \\ f(0) = f(1) = 0. \end{aligned} \tag{12}$$

Our diagonal catalysts differ from the transverse XX catalysts in Refs.[20,22]. Annealing-path choices can change perturbative crossings even with a fixed final problem[42]. They are temporary interactions, not a catalytic quantum state whose marginal must return unchanged. FEM does not simulate unitary quantum evolution.

Dense balance couplings and collective couplings, together with encoding auxiliaries, impose quantum connectivity and precision costs that the sparse classical implementation does not remove.

Coherent simulations

Eight synthetic hypergraphs use six vertices and four hyperedges of sizes 2, 3, 4 and 5, with fixed generation seeds 20–27. They require 14 qubits under Eq. (4). Full complex128 statevectors start in $|+\rangle^{\otimes 14}$. We simulate $\widehat{\mathbf{H}}(s) = -(1-s)\sum_j \widehat{\mathbf{X}}_j + [s\widehat{\mathbf{E}}_{\mathrm{QUBO}} + 4s(1-s)\widehat{\mathbf{C}}]/\lambda$, with durations 4 and 12 in units $\hbar = 1$. The diagonal catalyst uses pair strength 1.3 and auxiliary strength 0.55. Uncatalyzed runs set its envelope to zero. Midpoint Strang splitting alternates half diagonal phases, exact commuting single-qubit $\widehat{\mathbf{X}}$ rotations, and half diagonal phases, with time step at most 0.05. The first two instances are repeated at 0.025 for convergence checks. Exact enumeration supplies the optimal balanced cut and full-QUBO ground subspace. We report their total probabilities, rather than selecting one measurement or inferring quantum runtime from classical simulation time.

Dataflow workload

The application freezes three graphs, both raw fixed-policy base seeds (3,4), the strict KaHyPar partition, widths 8 and 32, 16 measured steps, three warmup steps and three repeats before outcomes. Trial order is shuffled with seed 27824. Original input and partition hashes are retained; the application partitions are not supplied back to the QUBO search. All runs start from the same graph/payload-specific numerical state.

Let $\mathbf{X}$ be vertex states, $\mathbf{B}$ incidence, $\mathbf{r}$ net sizes and $\mathbf{d}_{\mathrm{inc}} = (D_1, \ldots, D_n)^{\mathsf{T}}$ vertex incidence degrees. Each step evaluates

$$\begin{aligned} \mathbf{Y} &= \mathrm{diag}(\mathbf{r})^{-1}\mathbf{B}\mathbf{X}, \\ \mathbf{G} &= \mathrm{diag}(\max(\mathbf{d}_{\mathrm{inc}}, 1))^{-1}\mathbf{B}^{\mathsf{T}}\mathbf{Y}, \\ \mathbf{X}' &= \tanh(0.85\mathbf{X} + 0.15\mathbf{G} + \mathbf{Z}). \end{aligned} \tag{13}$$

The bias $\mathbf{Z}$ and initial states use a fixed seeded normal distribution. Each worker computes local net reductions. Only crossing-net vectors are sent to the other worker through a bidirectional process pipe; uncut nets require no payload. The two contributions produce the same net means as a serial reference. Actual sent/received buffer lengths verify $2d\times 8$ transmitted bytes per crossing net per step, excluding message headers. Received bytes equal transmitted bytes. Maximum absolute error to the serial calculation is recorded for every trial.

The experiment uses two local CPU processes, each pinned to one available logical CPU with numerical-library threads limited to one. Reported loop time is the larger worker elapsed time, excluding graph construction, process startup, warmup and final-state collection. Saved cuts and original partitioning times are reported alongside application runtimes. Runtime medians use three repeats; ranges and both seed results remain in the Supplementary Information. No confidence interval or network/FPGA speedup is inferred from these local measurements.

Benchmark protocol

Overlap-gap scope

For a fixed cut threshold τ, define the feasible low-cost set $\mathcal{S}_\tau = \{\mathbf{x} : \sum_i x_i = b,\ E_{\mathrm{cut}}(\mathbf{x}) \le \tau\}$. The complement-invariant vertex overlap is $q(\mathbf{x}, \mathbf{y}) = |n^{-1}\sum_i (2x_i - 1)(2y_i - 1)|$. An overlap-gap statement requires a forbidden interval of such overlaps between appropriate low-cost states, with its ensemble and threshold specified. The present real-graph runs and six-vertex coherent examples do not establish that property. Nor does a catalyst automatically evade an algorithmic obstruction: the resulting algorithm may still belong to a class for which a stability-based theorem applies. The curvature analysis concerns local variational stability only.

Endpoints and timing

The primary comparison has 30 graphs, three allowed balance settings, base seed 21 and 12,144 updates per 48-replica batch. The 1% study reuses the frozen graph, field and catalyst policy, with 64 central-safeguarded iterations specified for *scircuit* before observing its new batch. Numerical failures, if any, are retained and charged before a single safeguarded retry. Its KaHyPar references use seed 0 and a nominal 64 s limit; up to four independent single-thread reference processes run concurrently. The implicit balance solver uses 24

iterations, except for the 3% *scircuit* retry, which uses 64 with a central-bracket safeguard after both 24- and 64-iteration Newton attempts fail; its tolerance is unchanged. The original 20-graph policy and its ten-graph extension were fixed before their respective outcomes. The 3% slack-QUBO protocol was fixed before its outcomes. A preprocessing backend exception on *scircuit* was recorded before its first quality run: component-deflated, Jacobi-preconditioned CuPy LOBPCG replaces the stalled unscaled CPU Lanczos solve on the same incidence operator. Mode residuals, balance and orthogonality are independently checked. All attempted construction costs are retained. No graph name, KaHyPar partition or classical spectral bisection determines an initial state.

The three settings add 4,320 trajectories to 3,360 historical benchmark trajectories, giving 7,680 before the separate 480-trajectory structural controls. Historical budgets, seeds, failed outputs and exploratory policies are recorded in Supplementary Information. Raw outputs require zero auxiliary gap and the specified integer vertex-count bound. Each replica retains its minimum original-QUBO checkpoint energy using field signs and auxiliary-only minimization. Original-QUBO one-bit descent is then applied to every replica, without vertex projection or a feasibility filter on proposed moves. The strict *scircuit* winner is the only changed strict endpoint. Initial fields and complete binary states are independently reconstructed and scored. The 100 retained large-instance KaHyPar partitions are checked against input hashes, native cuts and block sizes. Classical mode rounding adds 240 balanced references to 16 historical references without additional FEM search.

FEM runs in FP64 on an NVIDIA A100-PCIE-40GB with manual gradients, grouped CSR products, compiled dense kernels and 32-update CUDA Graph replay. Single-update and 32-update capture agree in direct GPU tests. Batch time includes all 48 replicas and checkpoints. Charged pipeline time adds both single-thread CPU mode-construction costs, graph setup, compilation/capture and all-replica descent. The two CPU jobs can overlap; their sum is a conservative charged cost, not the measured parallel critical path. Component times are retained in a separate ledger.

The separate timing attribution, direct-encoding and mixed-timing controls contain 384, 576 and 288 trajectories, respectively. Their complete raw and selected states pass 2,496 independent score checks, including original-polynomial one-bit minima and matched initial-field hashes. These seen-instance controls are not counted in the primary 4,320 trajectories or used to replace its outcomes.

KaHyPar runs on one Intel Core i7-12700 thread. Each cut retains its actual process time and original 0%, 1% or 3% setting, including outputs that happen to be exactly balanced at 3%. KaHyPar and KaHyPar-E configurations and soft nominal limits of 64 or 128 s are recorded per reference. No unmeasured optimum, equal-hardware speedup or equal-budget advantage is inferred. Separate CUDA-event profiling and kernel traces are diagnostic measurements and never replace benchmark wall time.

Data availability

Input hashes, complete binary states, failed outputs, frozen protocols and independent scoring records are retained in the private research archive. Aggregate cut, observed-imbalance and timing records, input hashes and chart-reproduction scripts are distributed with the public SDK; complete benchmark states and failed outputs remain in the private archive. Benchmark provenance includes the ISPD98 netlists and the public KaHyPar benchmark collection[43], including sparse-matrix-derived instances from the University of Florida collection[44]. A permanent release and data-archive identifier remain to be assigned before submission.

Code availability

The project is maintained at https://github.com/yao-baijian/qubo-hypergraph-partition. A clean-history public SDK provides input handling, QUBO construction, independent scoring and a portable catalyst-off base FEM under a noncommercial licence. This reference retains the auxiliary Gibbs blocks, correlated slack block, implicit balance update, late penalty ramp and per-replica checkpoints. Independent mechanism tests and comparisons with small research trajectories verify these equations. The older independent- Bernoulli

Adam solver is a separate educational mode. The CPU reference does not implement catalyst construction or the benchmark's GPU acceleration. A platform-specific compiled catalyst backend is validated separately on seven paired configurations and does not yet reproduce the full 30-instance performance protocol. Development source, frozen source hashes and numerical audit records are retained privately. Executable reproduction artifacts and source access for peer review will be specified before submission.

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